

Brindlewythe Biosciences Ltd

BP16 protease programme: hERG channel screen

Final signed study report — revision B

Study BW03_0098

Accession B-ACC-2003-01676

Client: Belquorin Pharma

Date: 23 May 2003

Author: Dr B. Rowcroft, Screening Operations Director

1 Summary and control acceptance

The hERG channel screen covered 32 compounds. E-4031 gave a control IC50 of 13 nM and the compound median was 14 µM. Cell-level estimates, censoring and recording qualifications are retained in the released table.

Control acceptance preceded interpretation. Ines Alvarado checked the acceptance record and its link to the sample cards before the screening conclusions were released. Censored estimates and individual qualifications remain part of the accepted endpoint record; study acceptance does not remove them.

Table 1. Study summary and benchmarks

Metric	Value
Project	BW03_0098
Client	Belquorin Pharma
Compounds tested	32
Series / chemotype	Belquorin oxazole carboxamide series 16 (Oxazole carboxamide)
Assays	hERG patch clamp
Turnaround (calendar days)	16
Price	GBP 8,500
hERG IC50 <1 µM	0
hERG IC50 1-10 µM	14
hERG IC50 ≥10 µM	18
hERG median IC50 (µM)	14
E-4031 control IC50 (nM)	13

2 Study objective and chronology

Assess hERG current inhibition in the submitted screening set and provide a cardiac counterscreen profile for the chemistry decision.

The study covered 32 compounds for Belquorin Pharma under BW03_0098, accession B-ACC-2003-01676. Work began on 07 May 2003. The interim position was recorded on 15 May 2003 and the completed assessment was issued on 23 May 2003. Revision B supersedes the interim position.

The screen gate concerned the next chemistry decision. It was not a clinical assessment or an assertion that the intended target mechanism had been demonstrated. Dr B. Rowcroft was responsible for the scientific interpretation and final release.

3 Test articles and accountability

The full BP16 identifiers were retained from sample cards to run sheets. The project title describes a manual hERG patch-clamp assessment, while the supplied result table is labelled automated patch clamp. This report uses the common house endpoint name hERG channel screen and retains that method-label distinction with the study record rather than silently resolving it.

Test lot identity was preserved from receipt through preparation and measurement. The reportable compound identifier belongs to the material submitted for this study. A replacement lot, new solid form or formulation is not covered merely because it has the same chemistry designation.

The sample card provides the connection between retained material and the assay record. Follow-up work should carry that connection forward while recording any change of lot. Result transfer must use full identifiers rather than rack position or abbreviated suffix.

4 Experimental procedure

hERG current was measured by patch clamp with a repeated voltage protocol. A stable vehicle baseline preceded compound addition. Tail-current amplitude was normalised to the baseline in the same cell, and each concentration was allowed to approach a stable response before further addition. Current stability and recording quality were assessed before fitting.

The test concentration range extended to 30 μM . Independent cell recordings were retained in the endpoint table. Vehicle measurements addressed current drift; E-4031 demonstrated reference blocker sensitivity. The reference control is expressed in nM, while compound IC50 values are expressed in μM .

The run sheet retained baseline behaviour and recording limitations. An unacceptable recording was not rescued by averaging with acceptable cells. Available recordings and finite estimates are distinguished in the table. Absence of a fitted estimate at the top concentration is handled by censoring rather than substitution of that concentration as an exact IC50.

5 Plate acceptance and calculation

Dr R. Deshmukh checked reference sensitivity and current stability before interpreting the compound data. E-4031 gave an IC₅₀ of 13 nM. The study was accepted against the established recording and control criteria; individual cell or compound qualifications remained attached to the endpoints.

Normalised current inhibition was used for concentration-response fitting. Finite cell estimates were summarised geometrically where appropriate. Greater-than values were retained with their censoring symbols. The n cells (finite) field distinguishes available recordings from the subset contributing finite estimates and must accompany the compound summary.

Ines Alvarado checked units, column mapping and retention of qualifiers during report preparation. Bethany Rowcroft then considered the accepted endpoints against the study objective. A reference control demonstrates assay responsiveness under the test conditions; it does not establish specificity or suitability of a test compound outside those conditions.

6 Results: numerical record

The distribution was 0 compounds below 1 μ M, 14 in the 1–10 μ M band and 18 at or above 10 μ M. The supplied record contains 5 censored summaries and 2 flagged summaries. Censored compounds remain part of the distribution, not additional test articles.

Table 2. hERG channel screen: cell IC₅₀, compound summary and inhibition at 10 μ M

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
BP16-05400	>30	>30	>30	>30	21	3 (0)	
BP16-05406	1.4	3.0		2.1	86	2 (2)	
BP16-05412	26	24	>30	25	27	3 (2)	
BP16-05416	>30	20		>30	26	2 (1)	
BP16-05422	12	18	14	15	38	3 (3)	
BP16-05424	3.7	3.1		3.4	76	2 (2)	
BP16-05428	>30	>30		>30	25	2 (0)	First run rejected (E-4031 outside range); repeated
BP16-05430	3.6	1.8		2.6	79	2 (2)	
BP16-05434	6.1	13		9.0	49	2 (2)	
BP16-05440	12	17		14	39	2 (2)	
BP16-05442	5.1	5.6	6.8	5.8	61	3 (3)	
BP16-05446	9.4	6.6	4.6	6.6	66	3 (3)	
BP16-05448	11	5.6		7.7	50	2 (2)	
BP16-05460	30	>30		>30	32	2 (1)	
BP16-05478	15	19	9.6	14	39	3 (3)	
BP16-05482	13	21	14	16	40	3 (3)	First run rejected (E-4031 outside range); repeated
BP16-05484	9.3	15		12	42	2 (2)	
BP16-05494	18	23		20	39	2 (2)	
BP16-05496	9.1	4.9		6.7	60	2 (2)	
BP16-05502	7.7	6.8		7.2	55	2 (2)	
BP16-05508	14	13		14	40	2 (2)	
BP16-05518	12	16		14	37	2 (2)	
BP16-05520	17	17		17	42	2 (2)	
BP16-05532	20	11		15	38	2 (2)	
BP16-05536	21	24		23	27	2 (2)	
BP16-05538	>30	>30		>30	29	2 (0)	
BP16-05542	7.3	4.8		5.9	63	2 (2)	
BP16-05556	2.1	1.9		2.0	84	2 (2)	
BP16-05560	5.1	5.6		5.4	68	2 (2)	
BP16-05562	4.5	5.6		5.0	71	2 (2)	
BP16-05584	18	17		17	36	2 (2)	
BP16-05590	6.7	3.4	8.9	5.9	55	3 (3)	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 13 nM (acceptance 5–50 nM)

Compound IC₅₀ = geometric mean of finite cell IC₅₀s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)

The released tables are the primary numerical record. A blank cell is not a zero. A greater-than value is not an exact estimate. Concentration, matrix and endpoint units must remain attached to each result when it is copied from this report.

7 Comparison within the screening set

The lipophilic-cap branch was sufficiently concerning to change the chemistry direction. The absence of compounds in the below-1 μM band does not remove that screening concern. Less basic replacements were preferred rather than spending the next screen gate on further characterisation of the same branch.

Comparison within the present study is preferred to an unqualified comparison with historical screening sets. Test lot, preparation and assay conditions can change the apparent separation between candidates. A useful chemistry association remains a screening observation unless a further experiment distinguishes its possible causes.

The assay profile should be read in full before selecting a favourable isolated result. An optimisation preference does not change the endpoint, and a less favoured branch does not become an assay failure because the client chooses another direction.

8 Qualifications and interpretation limits

hERG current inhibition is an in vitro liability endpoint. It does not assess all cardiac ion channels, tissue response or clinical arrhythmia risk. A result above the tested range is not an exact potency estimate. No dose or exposure margin can be derived from these assay data alone.

Flagged summaries must be used with their QC notes. Report-level acceptance does not remove an individual qualification. A branch decision should not rely on a qualified estimate alone, and the released table retains the cell evidence needed to judge that distinction.

No exposure margin has been calculated. Such a calculation requires an agreed exposure measure and a justified connection between that measure and this assay endpoint. The report establishes neither a dose nor an in vivo response. These limits define the intended use of the screen.

9 Screen gate and client decision

The lipophilic cap carried an ion-channel liability, and Belquorin Pharma requested less basic replacements rather than further testing of that branch.

The next submission should address basicity and the cap region. Existing branch results remain reportable even though that branch is no longer preferred; abandoning a chemistry direction does not invalidate the assay.

Belquorin Pharma owns the chemistry decision. Brindlewythe Biosciences Ltd supplies the accepted experimental profile and its interpretation limits. The decision is recorded separately so that a progression preference is not mistaken for an additional experimental observation.

10 Follow-up work specification

Replacement compounds should receive their own hERG channel screen with a range that brackets the response where material permits. Retain reference-control logic and cell-level data. Exposure-based interpretation should follow only once the client has supplied an agreed pharmacokinetic context.

Any further study should state its question before allocating samples. Retain the existing compound identifier and test lot, and use a new accession for newly submitted material. A repeat intended to resolve a qualification should retain that purpose in its run sheet rather than being merged silently with the present result.

Proposed follow-up does not imply that the additional experiment has already been completed. Only the assays and endpoints in this report are released under the current study.

11 Study records and data transfer

Dr R. Deshmukh checked the electrophysiology assessment. Ines Alvarado checked transfer of cell values, unit labels and censoring symbols. Resolve recording qualifications through the original current records and run sheets rather than through a copied compound geomean.

The study file contains the sample-card link, preparation record, run sheets and released endpoint tables. Bethany Rowcroft prepared the interpretation and Ines Alvarado checked it against the accepted assay record. The final report should be retained with those records rather than used as a replacement for them.

When quoting the assessment, retain BW03_0098, revision B, the report date and endpoint units. A copied ranking without its qualifications loses information needed at the screen gate. This report is confidential to Belquorin Pharma and Brindlewythe Biosciences Ltd.

12 Conclusion and release

The lipophilic cap carried an ion-channel liability, and Belquorin Pharma requested less basic replacements rather than further testing of that branch.

The accepted experimental record supports that screening decision within the conditions specified. It does not extend the scope to untested compounds, matrices or biological endpoints. The numerical tables and qualifications remain the basis for any subsequent use of the results.

Revision B is the final report issued on 23 May 2003. The signatures record scientific release and the check of sample mapping, units and result qualifications.

Name	Role	Signature	Date
Bethany Rowcroft	Screening Operations Director — scientific release	<u>B Rowcroft</u>	23 May 2003
Ines Alvarado	Assay Quality Coordinator — report check	<u>I Alvarado</u>	23 May 2003